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The health care professional multimedia workstation: development and integration issues

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Abstract

Workstations for health care professionals provide access to distributed healthcare information systems. They are complex systems that have to manage, in a unified framework, the distributed and multimedia information needed for optimal patient care. The objective is to provide the end-user with a virtual and unified device that conceals the complexity of the underlying information system. Integration and reuse of modular software components are important in the development of such a workstation. A reference set of evaluation criteria is proposed that includes functional, technical, organizational, medical, cultural and ethical, economic and industrial components.

Key words: Healthcare professional workstation; Computerized patient record; Software engineering; Integration; Reuse; Technology evaluation

1. Introduction

Health care professionals depend on accurate and timely access to information — information about the patients under their care, information contained in the biomedical literature, and information accumulating in databases around the world. However, the promise of the information age has not been realized in health care because of the technical short-sightedness of closed systems and sociopolitical boundaries that inhibit collaboration and data sharing. Because of continued purchasing of closed-end solutions, health care facilities are still not in a position to provide their professionals with transparent access to distributed patient data, knowledge sources, and processing services. The workstation for the health care professional provides both a short-term solution to the problem of inadequate information infrastructures and an evolutionary path toward an integrated information environment [1–3].

The development of such a workstation involves consideration from the viewpoints of the clinician, the health administrator, and the engineer. From the clinician's perspective, the workstation can be more than a data and knowledge integrator. Through evolutionary steps, the workstation will first serve the information needs for patient care, support professional and personal

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communications, provide decision support, and facilitate professional and personal task management [4–8]. The workstation needs to evolve from an information display device to a work-facilitating device with the goal of becoming an intelligent electronic assistant.

From the health administrator's perspective, the workstation is a new technology to be integrated into the organization. This technology, however, is nothing less than a radical structural agent of change. The device can enhance productivity, change communication patterns and hence the ways in which professionals and support staff collaborate in the care process, provide the technical ingredients for true continuous quality improvement, and facilitate the collection of the data necessary for executive decision support.

Finally, from the engineer's perspective, the workstation is a hardware and software framework that needs to address interface management, storage, communication, processing and control. Physicially, this processes can be combined within the same box or distributed. The intelligent workstation is a virtual device which architecture should rely on accepted reference models, and development benefit from appropriate software engineering approaches [2,9,10].

Despite the fact that health care professionals have acquired considerable experience through dumb terminals or microcomputers with access to hospital information systems or to scientific or educational networks, the intelligent workstation is still at a conceptual or prototypical level (i.e. it is something to design, implement, and test in a near future). It raises the problems of functional specifications, development, integration, and evaluation (Fig. 1).

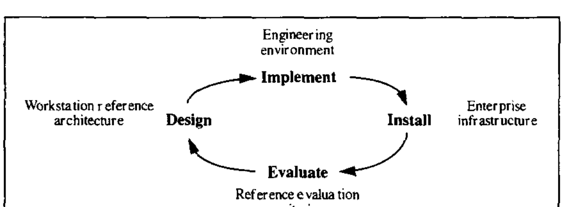

Fig. 1. The healthcare professional workstation development cycle.

2. Functional requirements

It is traditional in health care to assume that the functions of a computerized system should be patient-centered and the development user-centered. Interfaces and accessible functions need to be tailored to the needs of individual users or categories of users (e.g. physicians, nurses). Indeed, the participation of end-users in the development process and in evaluation teams with designers and external evaluators minimizes the risk of inappropriate design and provides the necessary feedback on the viability of the first prototypes [11].

But end-users' reactions are often biased by their personal experiences with products of previous generations. Requirements must also be based on the prospective understanding of the changes in the delivery of health care and in the technology. Comments on a prototype are different from experiments in a real situation. Such experiments require mature and secure products, and it becomes more difficult at this stage to make drastic changes. Economic pressures impose the need for a large market to balance the high cost of development. End-users' expectations are likely to be tempered by the constraint of minimal consensual functions.

Table 1, derived from the literature and the personal experience of the authors, summaries some key functional requirements of an intelligent workstation for the health professional [1,4–9]. Group 1 and part of group 5 functions are clearly patient-centered. Groups 3 and 4 indicate that direct help to the end-users' activity is a necessary condition of acceptability and success. Group 2 functions concern both the end-users and their environment (e.g. the institution).

3. Development tools

Design and implementation of an intelligent workstation are growing processes in which new functions are gradually added to existing ones. Changeability and adaptability are essential. For the workstation developer, the long-term objective is to conceive a virtual environment that gives the end-user the illusion of a single and unique appli-

Table 1
Selected functional requirements

Functional requirement
1. Ability to help the medical practice (direct patient care support)
Computer-based patient record
Data recording (administrative, medical, nursing, investigations)
Multimedia object management (e.g. text, graphics, images, sounds)
Data presentation (e.g. summaries, flow sheets, graphics)
Simultaneous record consultation (group work, staff discussions, managed care)
Decision support
Diagnostic and therapeutic suggestions
Strategy selection support
Clinical calculations (e.g. drug doses)
Search for similar patient cases
Access to contextual knowledge
Prognostic evaluation
Follow-up support (e.g. monitoring and alerts, follow-up reminders, compliance evaluation)
2. Support to the management of the health care professional environment
Healthcare unit management (resources, planning)
Access to reference information (e.g. protocols)
Evaluation of medical procedures (e.g. outcome measures)
3. Teaching and research
Access to literature and external databanks
Access to libraries of medical procedures
End-user assistance (e.g. on-line help, tutorials)
4. Logistic capabilities
Desktop publishing (e.g. text processing, spreadsheets, presentation tools)
Communication facilities (e.g. mailing, file/record transmission)
Personal data management (e.g. agenda, addresses)
Personalisation, customization of the workstation
5. Protection, security
End-user identification and authentication
Data integrity and security management
Maintenance of patient data privacy and confidentiality

cation, despite the fact that the functions rely on a set of distributed and heterogeneous pieces of hardware and software. Development is facilitated if the workstation software is initially conceived in a modular client/server design in which each component conforms to a normalized interface protocol [9,10,12]. The Domain-Specific Software Architecture (DSSA) methodology is an example of a method of producing specifications that permit the assembly of specialized components for a particular domain [13]. The spiral model of development enhances changeability and adaptability. At each step of a cycle, object-oriented methods facilitate modularity, encapsulation and component reuse.

Data and knowledge management systems can be used to represent the different classes of objects to be handled by the workstation. They include local or foreign objects such as those found in the hospital information system databases (e.g. patient administrative data, lab results), simple or complex objects (e.g. multimedia), static or dynamic objects such as actions and procedures (e.g. protocols or decision making processes). This last point leads to the construction of active databases that contrast with traditional static data repositories. Object-oriented database management systems can facilitate the constitution of metamodels and the management of distributed data dictionaries [9,12,14]. In the approach proposed by the Object Management Group, the information system acts as an interface repository that maintains the set of objects that are accessible through normalized interface definitions [15]. The repository stores the location of the foreign objects as well as the operations available for a given object and the means to access them.

Graphic user-interface management systems must allow the medical application developers to create end-user interfaces that are adapted to their professional habits and comply with medical style guides [16]. Indeed, end-users must be able to move between components that share a common look and feel in order to minimize the component training period and enhance acceptance of the workstation. Ethnographic studies of end-users' work practices (e.g. through video recording) might provide a better understanding of the cognitive processes involved in human-machine interaction, the barriers encountered, and the effect of education and training [17]. Health professionals are frequently interrupted in their activities. It is thus very important to be able to interrupt a transaction, to record the state of the interface when the interruption occurred, and then to be able to resume the stored configuration. Communication between professionals (within or between institutions) and cooperative work (groupware) are essential to modern care. Simultaneous consultation of the patient record and distance interaction constitute the foundation of such cooperative work.

Medical information must be transparently accessible whatever its nature or location. Thus, the workstation communication functions must allow the access to data or knowledge servers located within or outside the institution. Besides performing the necessary translation between different types of information presentation, the communication tools have to guarantee the semantic mapping between objects that may have been modeled differently [18,19]. From the standpoint of implementation, the communication tools must comply with e-mail and medical message exchange standards acting above communication and security standards [20].

Development environments have to deal both with the integration of components specifically designed and implemented for the use in a medical workstation, and with the encapsulation of external components that were not initially designed to cooperate (e.g. a text editor, a spreadsheet, a decision support system). The so-called common object request broker architecture (CORBA) allows objects of different origins to interact with one another and thus improve software modularity [15]. Indeed, as long as an object can issue a request in a standard format, called for by a standardized object request broker, objects can communicate transparently across different languages and platforms.

4. Installation constraints

The workstation needs to become part of the enterprise information system to facilitate the collaborative work of health care professionals, to increase quality and performances, and to allow growth and changes at the organizational level. Installation of an intelligent workstation is therefore part of the general strategy of the enterprise. At an institutional level, this implies [2,21]:

- the clear definition of a migration strategy;
- the selection of an appropriate network infrastructure supporting wide-area connectivity;
- the installation of applications that support standardized application programming interfaces (API) and system-independent repositories (client/server applications);
- an institutional commitment to promote and respect standards;
- procedures to check the conformity of the application to selected norms and standards, and to insure quality; and
- the organizational and financial investments needed to provide for the necessary training of the end-users, to promote collaborative work, and to create a dynamic for change.

5. Evaluation criteria

Establishment of a reference list of evaluation criteria might be relevant in different situations [22,23]. During the development process, a list of criteria can serve as internal validation, for example to avoid deviations from intended objectives

Table 2
Categories of criteria

Categories of criteria	Main actors	Questions
1. Functional Coverage Adaptability,...	Health care professionals	What?
2. Technical *Architectural* Extensibility, evolutivity, maintainability,... Modularity, distributivity, portability, openness (e.g. standards), interoperability, reusability,... Degree of integration (presentation, communication, data/knowledge, control),... *Quality and performances* Acceptability, ease of use, time response, reliability, security,...	Engineers	How?
3. Organizational End-users' participation Support for group decisions Changes in the relationships between actors,...	Decision makers	Which consequences?
4. Medical Quality of care Medical efficacy, effectiveness,...	Patients, health professionals	Improved quality of care?
5. Cultural and ethical Educational and research value Auditability, patients'/professionals'privacy,...	Society	Which values?
6. Economic Direct and indirect costs Return on investments,...	Decision makers	Which resources?
7. Industrial and commercial Industrial commitment, investments Market identification, market share,...	Industry	Which perspectives?

[11]. In this case, the list should be defined a priori and internal validation should be completed by independent evaluators. At a management level, evaluation can help with selection of the most appropriate workstation strategy (from a comparative study of scenarios) and with determining the impact of the installation of workstations in a given organization (e.g. cost-effectiveness evaluation). Intuitive perceptions of increased productivity, as claimed by vendors or perceived by end-users, have to be balanced with direct measurements. This is particularly difficult since counterproductivity may be expected with the introduction of a new tool and the learning curve may differ with different workstations or categories of users.

Table 2, adapted from a previous study [22], gives some examples of criteria relevant to the evaluation of a workstation for the health professional. The extent to which functions are covered (what the workstation does) and integrated (functional integration) is of immediate concern for the end-users.

At a technical level (i.e. how the workstation functions are implemented), conforming with software engineering principles (e.g. modularity, reusability, modularity) are important to guarantee accepted and perennial applications. Integration relates to the way the different components of the workstation constitute a coherent environment and can be considered around four dimensions [9]:

- presentation integration: do the different tools have a common appearance and behavior?
- data and knowledge integration: are the different medical objects taken into account? How do the tools share information? Is semantic integration achieved, and how?
- communication integration: how do the tools exchange information? Is the semantics of messages taken into account?
- process-control integration: is collaboration between the workstation tools achieved, and if so, how?

Organizational criteria should be distinguished when one is analyzing the result of a workstation experience and what can be attributed to the tool and to its environment. Similar complex interrelationships obviously appear for medical, cultural or ethical criteria.

6. Discussion and conclusion

The multimedia workstation for health care professionals, integrated in a network architecture, is the natural bridge to distributed and integrated health information systems and offers the support for telemedicine and collaborative work in the health area. Medical decisions rely on consultation of the patient record, analysis of similar cases, and access to contextual knowledge. The workstation allows the end-user to view the patient record as a single entity when the information is distributed among different servers. Automatic coupling with knowledge servers facilitates the search for the associated knowledge, and standardization of query languages facilitates the access to similar patient cases from existing databases.

Part of the difficulty in the development of an intelligent workstation is related to the management of multimedia medical objects that are expressed with multiple formalisms and reside on heterogeneous machines. This requires the use of collections of dedicated tools and their integration in the common workstation platform. Reuse of existing software components is essential, since major elements already exist in the marketplace and cannot be rebuilt from scratch. But integration and reuse might be conflicting goals [10]. For example, the success of office integrated products (e.g. combination of a word processor, a worksheet, a graphics system, and a database) is in contradiction with modularity, which is a condition for reuse. Constitution of a meta-database (e.g. through an object-oriented model) facilitates the transparent access to heterogeneous information repositories and semantic integration [18,19] but limits the dynamic inclusion of new components (i.e. new data structures have to be declared in the meta-database to be transparently accessed).

Another difficulty lies in the fact that an intelligent workstation is something to add to an existing informations system, acting as a mediator between the end-user and the underlying information system. For an institution, installation of intelligent workstations is different from the purchase of a set of micro-computers or graphics stations. Decision makers should be convinced for the necessity of a clear migration strategy and conscious of the importance of the financial, educational and organizational investments that are required. Experiences with collaborative development of workstation components and promotion of reusable software libraries are two areas where substantial progresses might be expected.

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